SOLAR PV • AC / HVAC INFRASTRUCTURE | CLOSED-LOOP AI COPILOT | MANDVI • KUTCH PILOT GROUNDED

The Operational AI Copilot for Physical Field Service Enterprises

Eliminating the 32% first-visit failure rate in high-voltage Solar PV and critical HVAC operations through grounded vernacular AI intake, 5-factor deterministic dispatch, and mandatory physical telemetry benchmarking.

94%	Zero	4.6×
FIRST-TIME FIX RATE	WASTED TRUCK ROLLS	MONTHLY CONTRACTOR ROI

Founder & CEO: Faiz Khatri • Gujarat, India
Live 3-Min Video Walkthrough: `SERVORA_3MIN_YC_DEMO.mp4`

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The Physical Service Crisis in High-Voltage Infrastructure

Solar EPC and commercial HVAC contractors lose up to 35% of their net margins on avoidable return visits.

BOTTLENECK 1: BLIND INTAKE

Vernacular Ambiguity

Customers report vague vernacular complaints like *"Solar generation stopped since morning"* or *"inverter red light blinking"*. Dispatchers lack engineering tools to diagnose root causes before sending technicians.

OPERATIONAL CONSEQUENCE:
Technicians arrive on site blind without knowing whether the fault is inverter, surge, grid, or panel.

BOTTLENECK 2: MISSING SPARES

32% First-Visit Failures

Because the root cause was unknown, the technician didn't bring the required 1000V DC surge cartridge or MC4 crimping tool. The job stalls, requiring a second truck roll days later.

DIRECT FINANCIAL LOSS:
Average cost of a wasted truck roll: **₹3,500 – ₹5,000 ($45–$65)** in labor, fuel, and SLA penalty.

BOTTLENECK 3: RANDOM DISPATCH

Unskilled Assignment

Legacy dispatch tools match solely on GPS proximity. They send a junior apprentice to a 10kW high-voltage solar string inverter issue, creating severe safety risks and warranty voidance.

THE AI HALLUCINATION FEAR:
Generic LLM tools hallucinate dangerous repair guidance without grounding in physical electrical standards.

The Servora Closed-Loop Execution Platform

Not another CRM or generic ticketing tool. An operational intelligence system that connects physical field reality to deterministic dispatch.

PILLAR 1 • SMART INTAKE

Diagnostic Normalization

AI symptom normalizer ingests colloquial customer descriptions, extracting fault codes (e.g. `E-031 / DC Transient`) and system symptoms into structured engineering tickets.

- ✓ Location coverage verification within 25 km radius
- ✓ Predicts probable root cause (88% confidence)
- ✓ Recommends mandatory tools & spare cartridges

PILLAR 2 • SMART DISPATCH

5-Factor Match Engine

Algorithmic dispatch assigns the optimal technician based on deterministic mathematical weighting, not arbitrary geographic proximity.

- ✓ **Certification (40%)** : Inverter Level-2 vs HVAC
- ✓ **Tools Onboard (20%)** : Multimeter, clamp meter verified
- ✓ **Status (20%)** : Available route slot
- ✓ **Proximity (10%)** : Travel overhead cut by 34%
- ✓ **Experience (10%)** : Years on specific equipment types

PILLAR 3 • GROUND TRUTH

Physical Verification Moat

Technicians must record physical multimeter telemetry (Voc, Current, Thermals) and photo evidence. Servora benchmarks AI predictions against ground reality.

- ✓ Guarantees strict audit-compliant SLA timestamps
- ✓ Eliminates warranty fraud and billing disputes
- ✓ Retrains regional models with verified physical telemetry

The 3-Minute Live App Choreography (Mandvi, Kutch)

Recorded live without slides or mockups in `SERVORA_3MIN_YC_DEMO.mp4` across all 3 portals.

`ACT 1 • 00:00 - 00:55`

Customer Intake

Homeowner in Mandvi (PIN 370465) verifies 25 km service radius. Taps *"Inverter Offline / Fault Code"* with error `E-031`.

```
Ticket #SOL-2026-0084 created with AI urgency triage
```

`ACT 2 • 00:55 - 01:55`

Manager HQ Triage

Operations Manager inspects AI hypothesis: 88% probability DC surge burnout. 5-Factor match ranks Rahul Sharma #1 (96.4%). Dispatched in 1 click.

```
Spares checked: 1000V DC surge cartridge & multimeter
```

`ACT 3 • 01:55 - 02:40`

Field Execution

Tech mobile terminal advances: *Accepted → On The Way → On Site*. Rahul tests Voc = 382V, replaces cartridge, uploads before/after photos.

```
Multimeter logged: Voc 382V • Cartridge replaced
```

`ACT 4 • 02:40 - 03:00`

Audit & Closeout

Loop closes at HQ: AI Guess matches Physical Ground Fix (100% score). Digital customer signoff captured. Automated invoice and warranty card issued.

```
Total cycle: 42 mins (vs 120m target) • Zero disputes
```

VIDEO SPECS Exact 180.00s Duration • 1920×1080 Full HD @ 24fps • Size: 18.41 MB • H.264 Progressive

File location: `SERVORA_3MIN_YC_DEMO.mp4`

Designed for Field Realities Across 3 Interconnected Portals

High-contrast outdoor mobile terminals for technicians; comprehensive command center suites for operations managers.

Customer & Facility Portal

LOCATION-FIRST • ZERO FRICTION

Eliminates customer frustration by verifying local service coverage upfront before booking.

- ✓ Auto-detects PIN code & service zone
- ✓ Voice-to-text problem description
- ✓ Real-time technician ETA tracking & privacy masking
- ✓ Permanent digital asset warranty vault

Operations Command Center

HQ FLEET COMMAND • CENTRAL CONTROL

Command suite giving operations heads real-time visibility into every field asset and unit.

- ✓ Live Kanban ticket dispatch pipeline
- ✓ AI Pre-Arrival diagnostic package viewer
- ✓ Fleet radar map & dynamic technician routing
- ✓ Pilot Insights: AI vs Actual accuracy benchmarks

Field Engineer Mobile Terminal

HIGH-CONTRAST PWA • DIRECT SUNLIGHT

Engineered for physical technicians working on sunny rooftops and dusty plant rooms.

- ✓ Calibrated $ge 44 ext{px}$ touch targets for gloved fingers
- ✓ Guided electrical diagnostic trees & multimeter forms
- ✓ Mandatory multi-photo verification before closeout
- ✓ Works in intermittent network coverage with offline sync

The Ground-Truth AI Telemetry Moat

Every service visit closes an empirical loop: comparing AI hypothesis against physical multimeter readings.

The Closed-Loop Intelligence Flywheel

1. Pre-Visit AI Hypothesis:
Generates root-cause probability (88% DC Isolator / Surge cutoff) and required spare part list.

2. Physical Fieldwork Telemetry:
Technician records on-site multimeter readings ($V_{oc}=382 ext{V}$, $I_{sc}=8.4 ext{A}$) and confirms replaced components.

3. Ground-Truth Empirical Audit:
System scores AI Diagnostic Accuracy (0–100%) and logs discrepancies to retrain regional fault models.

Active Pilot Benchmarks (Live Pilot Engine)

91.4%
DIAGNOSTIC MATCH RATE
AI hypothesis matched physical field findings

89.8%
SPARES PREDICTION ACCURACY
Technicians had exact replacement cartridge on hand

42 min
AVERAGE TURNAROUND
SLA Target: 120 mins (78 mins saved per ticket)

100%
AUDIT SLA COMPLIANCE
Zero unverified repairs or billing disputes

Grounded Customer Discovery: Mandvi & Kutch Region

We didn't build in an ivory tower. We cataloged and validated 12 actual Solar & HVAC businesses in coastal Gujarat.

The Kutch Operational Reality

Mandvi and Kutch represent Gujarat's highest rooftop solar density per capita under regional solar initiatives, but suffer unique coastal operating challenges:

- ✓ **Severe Coastal Salt Corrosion** : High salinity causes frequent DC isolator cartridge burnout and copper oxidation on inverters within 18 months.
- ✓ **Critical Technician Scarcity** : Contractors employ general electricians who misdiagnose inverter firmware cutoffs as dead panels, wasting days.
- ✓ **Delayed Billing & Churn** : 30–60 day paperwork turnaround before contractors can collect service payments.

Verified Local Discovery Cohort

BUSINESS NAME	VERTICAL	LOCATION	FLEET SCALE
Shreeji Solar & Electricals	SOLAR EPC	Mandvi, Kutch	6 Field Technicians
Kutch Solar Tech	SOLAR EPC	Coastal Mandvi Belt	8 Field Technicians
Al-Madina Air Conditioning	AC / HVAC	Mandvi Bazaar	4 Field Technicians
Kutch Cooling Centre	AC / HVAC	Ghatkopar Road, Mandvi	5 Field Technicians

Full 12-contractor dossier compiled in `SERVORA_MANDVI_KUTCH_LEAD_DIRECTORY.pdf`

B2B SaaS Pricing & Compelling Contractor ROI

Contractor pays a predictable monthly subscription. Eliminating just 2 repeat truck rolls pays for the entire software 4× over.

TIER 1 • STARTER OPS

₹4,999 / month

($60/mo • Ideal for small regional contractors)

- ✓ Up to 5 Field Technicians
- ✓ Customer Intake & Territory Validation
- ✓ Digital Work Orders & Photo Evidence
- ✓ Mobile PWA for Field Technicians

TIER 2 • GROWTH FLEET — POPULAR

₹14,999 / month

($180/mo • Full AI Dispatch & Diagnostic Engine)

- ✓ Up to 20 Field Technicians
- ✓ **AI Symptom Normalizer & Root-Cause Pre-Triage**
- ✓ **5-Factor Intelligent Match Dispatch Engine**
- ✓ **Pilot Insights Ground-Truth Benchmarking**
- ✓ Automated Digital Invoicing & SLA Dashboards

TIER 3 • ENTERPRISE EPC

₹39,999 / month

($480/mo • Multi-Branch & OEM Integration)

- ✓ Unlimited Field Technicians & Branch Depots
- ✓ Direct OEM Telemetry API integration
- ✓ Custom Warranty Claim Automation
- ✓ Dedicated Enterprise SLA & Account Manager

Contractor ROI Equation (A 10-Tech Team):
Eliminating 2 repeat visits per tech/mo saves 20 visits × ₹3,500 = **₹70,000 saved monthly** vs ₹14,999 software cost.

4.6× Net ROI

Why Servora Wins: High-Voltage Vertical Specialization

Legacy desktop software doesn't understand inverters; generic gig portals commoditize technicians without solving operational failures.

OPERATIONAL CAPABILITY	SERVORA	LEGACY DESKTOP CRMS	GENERIC WORK ORDER TOOLS	CONSUMER GIG PORTALS
Domain AI Triage (Solar/HVAC)	NATIVE FAULT CODES	NO DOMAIN CONTEXT	GENERIC TEXT NOTES	FIXED PRICE MENUS ONLY
5-Factor Skill & Tool Dispatch	DETERMINISTIC ALGORITHMIC	MANUAL ASSIGNMENT	GPS DISTANCE ONLY	RANDOM POOL BIDDING
Physical Multimeter Telemetry	VOC, CURRENT, ARC-FLASH LOG	NONE	NONE	NONE
Ground-Truth AI Audit Loop	AI VS PHYSICAL BENCHMARK	NONE	NONE	NONE
Emerging Market Offline PWA	HIGH-CONTRAST DIRECT SUN	HEAVY DESKTOP FIRST	APP STORE NATIVE ONLY	CONSUMER CENTRIC

Milestone Roadmap & Enterprise Scaling

Scaling from verified Gujarat pilot contractors to global clean-energy and electrification field operations.

PHASE 1 • Q3–Q4 2026

Regional Pilot Scale

- ✓ Onboard 25 Solar EPC & HVAC Contractors across Kutch, Rajkot, and Ahmedabad.
- ✓ Process 1,500+ live verified field work orders.
- ✓ Achieve ₹15 Lakhs ($18,000) Annualized ARR.
- ✓ Lock in first OEM telemetry pilot partnership.

PHASE 2 • 2027

National Clean-Tech Expansion

- ✓ Expand across India's top 6 solar states (Gujarat, Rajasthan, Maharashtra, Karnataka).
- ✓ Automated inverter warranty claim processing directly with Tier-1 OEM manufacturers.
- ✓ Surpass ₹1.5 Crore ($180,000) ARR with 120+ active fleets.

PHASE 3 • 2027–2028

Global Sunbelt Electrification

- ✓ Launch US Sunbelt (California, Texas, Florida) and MENA rooftop solar operations.
- ✓ Expand vertical coverage into EV Fast Charging stations and Heat Pumps.
- ✓ Become the global standard OS for high-voltage distributed clean energy field service.

Faiz Khatri • Founder & CEO
Servora Technologies • Building the Future of Clean-Tech Operations

Full HD 3-Minute Video Demo Available
`SERVORA_3MIN_YC_DEMO.mp4` • Running live on `localhost:5173`